

PU-I Biology · Chapter 9

Biomolecules

Handwritten Notes — Narendra Sir · BAIO Learning System

1 Chemical Composition of Living Tissue

Are all organisms made of the same chemicals?

- Elemental analysis of plant / animal tissue / microbial paste gives C, H, O, N, S, Na, Ca, Mg ...
- Earth's crust gives a **similar list** → no element is unique to living matter
- But **C & H** are far more abundant in living tissue than in earth's crust ★ exam

Element	Earth's crust %	Human body %
Hydrogen (H)	0.14	0.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Silicon (Si)	27.7	negligible

Analysis of organic compounds — TCA method

1. Grind living tissue in **trichloroacetic acid (Cl_3CCOOH)** → slurry
2. Filter → **filtrate = acid-soluble** (biomolecules) + **retentate = acid-insoluble** (biomacromolecules)
3. Compounds in filtrate separated by separation techniques → formula & structure by analytical techniques

 **Remember:** Acid-soluble + acid-insoluble fractions = entire chemical composition of the cellular pool.

Analysis of inorganic compounds — Ashing

Dry tissue (→ dry weight) → **burn** → organic & gaseous compounds removed → **ash**.
Ash contains: Ca, Mg, S, P · ions Na^+ , K^+ , Ca^{2+} , Mg^{2+} · NaCl , CaCO_3 , PO_4^{3-} , SO_4^{2-}

What is a biomolecule?

Def ► All carbon compounds obtained from living tissues are called biomolecules.

- Chemistry view → by functional groups (aldehydes, ketones, aromatic compounds)
- Biology view → by size: **micromolecules** (< 1000 Da) & **macromolecules** (> 1000 Da)

Biomicromolecules (<1000 Da)

amino acids · fatty acids ·
monosaccharides · nitrogenous bases

Biomacromolecules (>1000 Da)

proteins · nucleic acids ·
polysaccharides (+ lipids*)

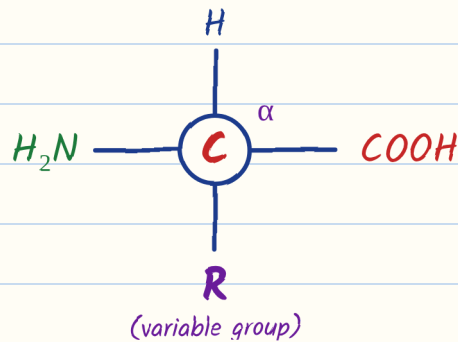
* **Lipids exception:** mol. wt. only ~800 Da, yet in the **acid-insoluble** fraction — they are not polymers; on grinding, cell membranes break into **water-insoluble vesicles** that come down with the pellet. ★ exam

Average composition of a cell

Component	% of total cellular mass
Water	70–90 (most abundant)
Proteins	10–15
Nucleic acids	5–7
Carbohydrates	3
Lipids	2
Ions	1

2 Amino Acids

Def ► Organic compounds with an **amino group** and an **acidic (carboxyl) group** on the same carbon — the **α -carbon** → hence **α -amino acids**.



General structure of an α -amino acid

- 21 types based on the R group (NCERT: proteins use 20) ★ exam
- $R = H \rightarrow$ glycine (simplest) · $R = CH_3 \rightarrow$ alanine · $R = CH_2OH \rightarrow$ serine

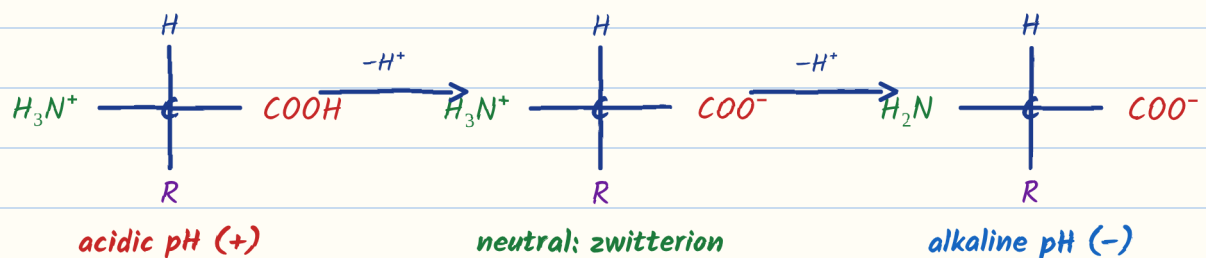
Classification (by number of $-NH_2$ & $-COOH$)

Type	Feature	Examples
Acidic	2 $-COOH$	glutamic acid, aspartic acid
Basic	2 $-NH_2$	lysine, arginine
Neutral	equal $-NH_2$ & $-COOH$	valine, glycine, alanine
Aromatic	benzene ring in R	tyrosine, phenylalanine, tryptophan

Essential vs non-essential

- **Essential** — cannot be made by the body → from diet: isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine (+ arginine, histidine conditionally)
- **Non-essential** — made by the body: alanine, asparagine, aspartic acid, cysteine, glutamic acid, glutamine, glycine, proline, serine, tyrosine

Zwitterion & isoelectric pH



Zwitterion = dipolar ion with both + and - charges (in neutral solution).

Isoelectric pH = pH at which the amino acid is electrically neutral. ★ exam

3 Lipids

Def ▶ Lipids are **water-insoluble** biomolecules — a fatty acid, glycerol, or both.

Fatty acids — R-COOH

- R = methyl ($-\text{CH}_3$), ethyl ($-\text{C}_2\text{H}_5$) or $-\text{CH}_2-$ chain of 1 to 19 C
- **Palmitic acid** — 16 C incl. carboxyl C · saturated · palm oil, butter, dairy
- **Arachidonic acid** — 20 C incl. carboxyl C · unsaturated · membrane phospholipids, brain



Saturated (no C=C) e.g. palmitic

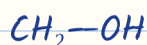
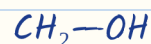


Unsaturated (C=C in red)

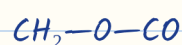
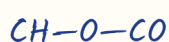
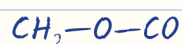
Palmitic acid = 16 C (incl. carboxyl C) · Arachidonic acid = 20 C

Glycerol & glycerides

Glycerol = trihydroxy propane (simple lipid). Fatty acids esterified to glycerol → mono-, di-, tri-glycerides.



Glycerol



Triglyceride (3 fatty acids esterified)



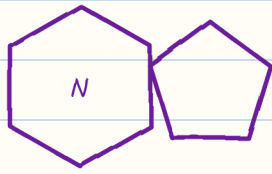
Fats — high melting point, solid
ghee, butter, vanaspati

Oils — low melting point, liquid in
winter
gingelly oil, coconut oil, palm oil

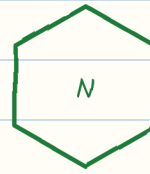
Phospholipids & others

- **Phospholipid** = lipid with phosphorus + phosphorylated organic compound → **lecithin** (cell membranes)
- Neural tissues have lipids with more complex structures
- **Sphingolipids** (brain) · **steroids** (testosterone, oestrogen, progesterone) · **sterols** (cholesterol, vitamin D) · **prostaglandins**

4 Nitrogenous Bases, Nucleosides & Nucleotides

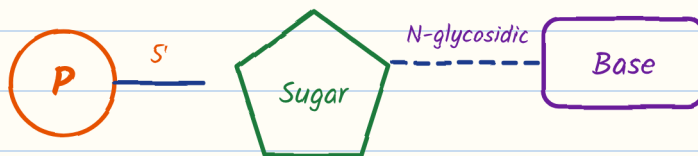


Purine (2 rings) A, G



Pyrimidine (1 ring) C, T, U

Base	Type	Occurs in	Nucleoside	Nucleotide
Adenine (A)	purine	DNA & RNA	adenosine	adenylic acid
Guanine (G)	purine	DNA & RNA	guanosine	guanylic acid
Cytosine (C)	pyrimidine	DNA & RNA	cytidine	cytidylic acid
Thymine (T)	pyrimidine	DNA only	thymidine	thymidylic acid
Uracil (U)	pyrimidine	RNA only	uridine	uridylic acid



Nucleotide = Base + Sugar + Phosphate

Nucleoside
= Base + Sugar

Nucleotide
= Nucleoside + P

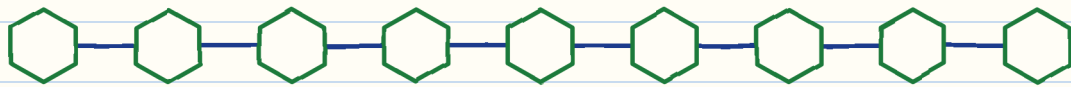
Remember: PUre As Gold → purines A, G · CUT the PY → pyrimidines C, U, T

5 Biomacromolecules & Polysaccharides

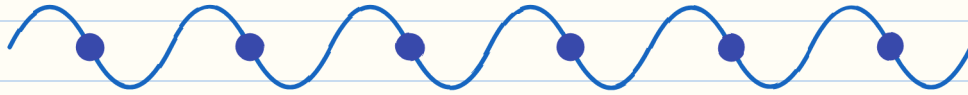
→ Acid-insoluble fraction: proteins, nucleic acids, polysaccharides, lipids → mol. wt. > 1000 Da (except lipids)

Def ▶ Polysaccharides = long chains of monosaccharides joined by glycosidic bonds.

- **Homopolysaccharide** — one type of monosaccharide (cellulose, starch, glycogen, inulin)
- **Heteropolysaccharide / complex** — amino-sugars & chemically modified sugars (chitin)



Cellulose: β -glucose, β -1,4, straight unbranched chain



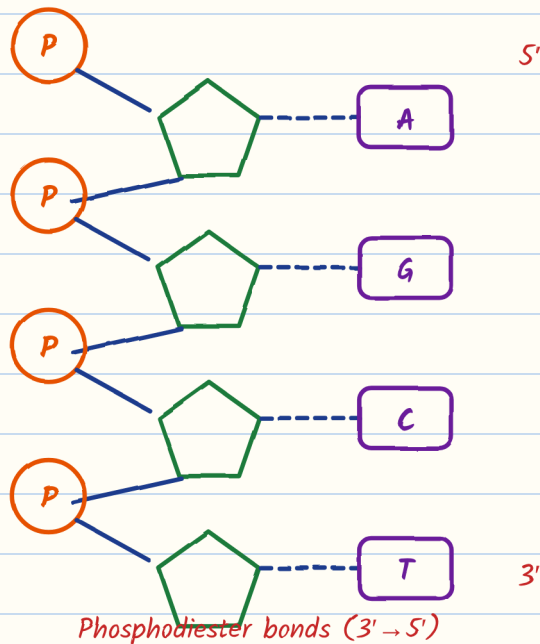
Starch helix holds I_2 (•) $\rightarrow$ BLUE colour

Polysaccharide	Monomer	Key points
Cellulose	β -glucose	β -1,4 · unbranched · plant cell wall, cotton, paper · no blue with I_2
Starch	α -glucose	α -1,4 · helical, branched · plant energy store · blue with I_2 ★ exam
Glycogen	glucose	animal starch · more branched than starch
Inulin	fructose	storage in dahlia, chicory roots
Chitin	amino-sugars	exoskeleton of arthropods, fungal cell wall

💡 **Remember:** In a polysaccharide chain the right end = reducing end, left end = non-reducing end.

6 Nucleic Acids

Def ▶ Nucleic acids = polynucleotides (polymers of nucleotides).



- Sugar: **ribose** → **RNA**, **2'-deoxyribose** → **DNA** (no O at 2')
- **Phosphodiester bond**: phosphate links the 3'-C of one sugar to the 5'-C of the next ★ **exam**
- Gives a repeating **sugar-phosphate backbone**; bases project from each sugar
- Base-sugar = **N-glycosidic bond**
- Properties (genetic information) depend on the **sequence of bases**

Feature	DNA	RNA
Sugar	2'-deoxyribose	ribose
Unique pyrimidine	thymine (T)	uracil (U)
Strands (usually)	double helix	single
Unit	deoxyribonucleotide	ribonucleotide

7 Proteins

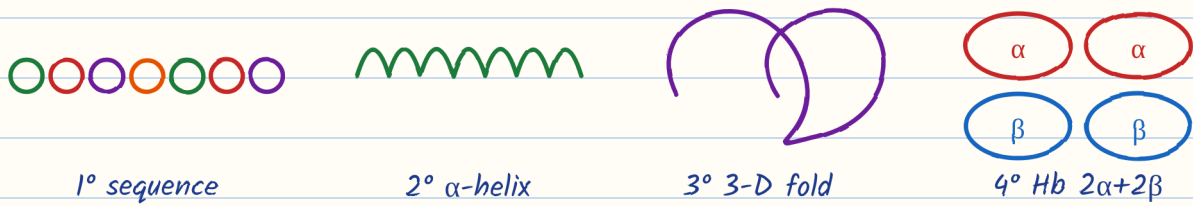
Def ▶ Proteins are **polypeptides** — linear chains of amino acids linked by **peptide bonds**. They are **heteropolymers**.

Functions

Protein	Function
Collagen	intercellular ground substance
Trypsin	enzyme
Insulin	hormone
Antibody	fights infectious agents
Receptor	sensory reception (smell, taste, hormone)
GLUT-4	glucose transport into cells

Collagen = most abundant protein in the **animal world / human body** · **RuBisCO** = most abundant protein in the **whole biosphere** ★ **exam**

Four levels of protein structure



Primary → Secondary → Tertiary (biologically active) → Quaternary

- **Primary** — sequence of amino acids; first = **N-terminal**, last = **C-terminal**
- **Secondary** — local folding: α-helix (only **right-handed** helices), β-pleated sheet; H-bonds
- **Tertiary** — 3-D folding like a hollow woollen ball → **biologically active**
- **Quaternary** — 2 or more subunits; adult **haemoglobin** = 2α + 2β ★ **exam**

Primary vs secondary metabolites

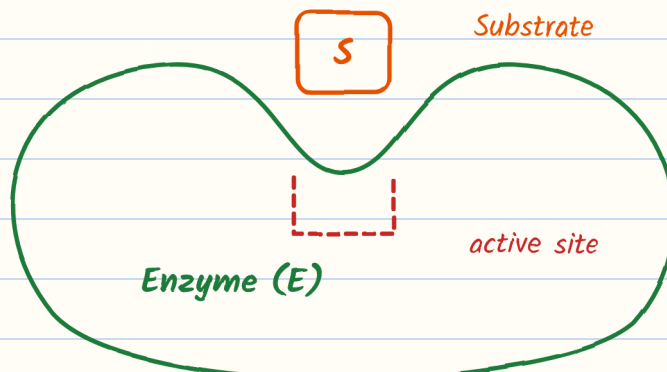
Primary — known roles in growth & physiology (amino acids, sugars, fatty acids).

Secondary — not directly in growth; useful to humans or ecologically important:

Pigments	carotenoids, anthocyanins	Toxins	abrin, ricin
Alkaloids	morphine, codeine	Lectins	concanavalin A
Terpenoids	monoterpenes, diterpenes	Drugs	vinblastin, curcumin
Essential oils	lemon grass oil	Polymers	rubber, gums, cellulose

8 Enzymes

Def ▶ Enzymes are biological catalysts — **almost all are proteins**. Exception: **ribozymes** (nucleic acids acting as enzymes).



Substrate fits into the active site (a crevice / pocket in the tertiary structure)

- Damaged at high temperature (above ~40°C) — unlike inorganic catalysts

→ Enzymes of **thermophilic** organisms (hot vents, sulphur springs) stable up to 80–90°C

Holoenzyme = apoenzyme + cofactor

Cofactor	Nature / binding	Example
Prosthetic group	organic, tightly bound to apoenzyme	haem in peroxidase, catalase
Co-enzyme	organic, transient association during catalysis	NAD, NADP (contain vitamin niacin)
Metal ion	coordination bonds with active site & substrate	Zn^{2+} in carboxypeptidase

 **Remember:** Removing the cofactor → the enzyme loses its catalytic activity.

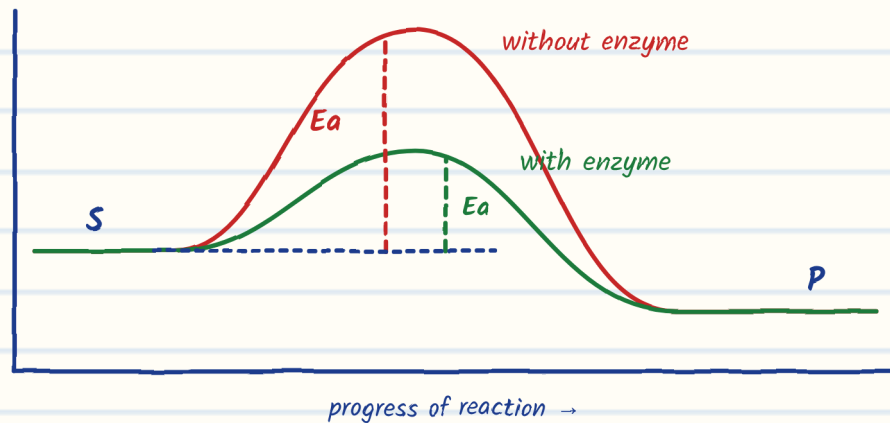
Six classes of enzymes — “OTHLIL” ★ exam

Class	Reaction
1. Oxidoreductases / dehydrogenases	$S \text{ reduced} + S' \text{ oxidised} \rightarrow S \text{ oxidised} + S' \text{ reduced}$
2. Transferases	transfer a group G (not H): $S-G + S' \rightarrow S + S'-G$
3. Hydrolases	hydrolysis of ester, ether, peptide, glycosidic, $C-C$ bonds
4. Lyases	remove groups by non-hydrolysis → double bonds
5. Isomerases	inter-conversion of optical, geometric or positional isomers
6. Ligases	link two compounds ($C-O$, $C-S$, $C-N$, $P-O$ bonds)

9 Enzyme Action & Factors

- Rate of reaction = $\delta P / \delta t$ · roughly doubles / halves for every 10°C change
- $CO_2 + H_2O \rightleftharpoons H_2CO_3$: ~200 molecules/hour without enzyme → ~600,000/second with carbonic anhydrase (~10 million times faster) ★ exam
- Glucose → 2 pyruvic acid: 10 enzyme-catalysed steps

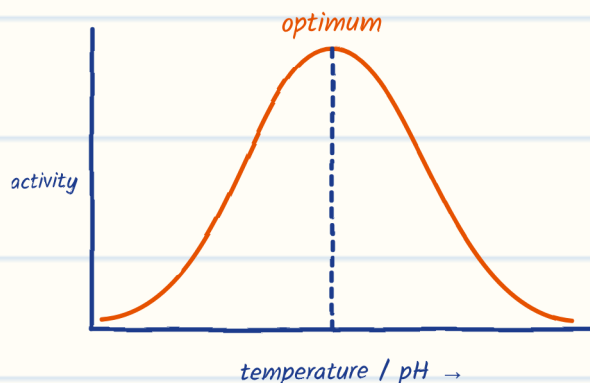
$E + S \rightleftharpoons ES \text{ complex} \rightarrow EP \rightarrow E + P$ · transition state = high-energy, unstable
· enzymes lower the activation energy (E_a)



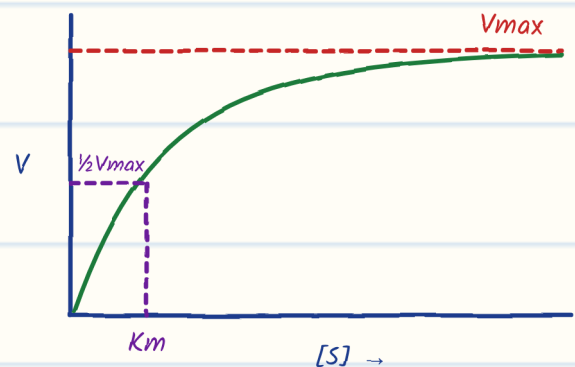
Catalytic cycle

1. Substrate binds the active site
2. Enzyme shape changes — fits tightly around substrate
3. Bonds of substrate break → **EP complex**
4. Product released → free enzyme ready again

Factors affecting enzyme activity



Temperature & pH — optimum



Substrate concentration

- Low temperature → enzyme **inactive** (reversible) · high temperature → **denatured**
- Velocity rises with $[S]$ then reaches V_{max} (enzyme saturated) · $K_m = [S]$ at $\frac{1}{2} V_{max}$
- **Competitive inhibitor** resembles the substrate → competes for active site ·
malonate inhibits succinic dehydrogenase ★ exam

Quick revision ✓

Lipids ~800 Da but acid-insoluble · Collagen (animals) / RuBisCO (biosphere) · Hb $2\alpha+2\beta$ · starch + I_2 = blue · inulin = fructose · phosphodiester $3' \rightarrow 5'$ · OTHLIL · malonate → succinic dehydrogenase · $K_m = [S]$ at $\frac{1}{2} V_{max}$